

Circle Theorems

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

What is the name of a straight line drawn from the centre of a circle to a point on the circumference?

- 1 It is half of a diameter.
- 2 Every radius of a circle is the same length.

Answer: a radius

2 [2 marks]

- (a) What is a chord of a circle?
- (b) What is a tangent to a circle?

- 1 **(a)** A chord is a straight line joining two points on the circumference. (A diameter is the special chord that passes through the centre.)
- 2 **(b)** A tangent is a straight line that touches the circle at exactly one point.

Answer: (a) a line joining two points on the circumference (b) a line touching the circle at exactly one point

3 [2 marks]

AB is a diameter of a circle. C is another point on the circle.

Write down the size of angle ACB and give a reason.

- 1 The angle standing on a diameter, drawn from the circumference, is always a right angle.
- 2 This is the 'angle in a semicircle' theorem.

Answer: 90° — the angle in a semicircle is a right angle

4 [2 marks]

A, B and C are points on a circle with centre O. Angle AOC = 100° and B lies on the major arc.

Work out angle ABC.

- 1 The angle at the centre is twice the angle at the circumference on the same arc.
- 2 $100 \div 2 = 50^\circ$

Answer: 50°

5

[3 marks]

PQRS is a cyclic quadrilateral. Angle PQR = 105° .

Work out angle PSR and give a reason.

- 1 PQR and PSR are opposite angles of a cyclic quadrilateral.
- 2 Opposite angles of a cyclic quadrilateral add up to 180° .
- 3 $180 - 105 = 75^\circ$

Answer: 75° — opposite angles of a cyclic quadrilateral add to 180°

6

[3 marks]

A tangent touches a circle of radius 5 cm at point T. P is a point on the tangent with PT = 12 cm. O is the centre.

Work out the length of OP.

- 1 A tangent meets a radius at 90° , so triangle OTP is right-angled at T.
- 2 $OP^2 = OT^2 + TP^2 = 5^2 + 12^2$
- 3 $= 25 + 144 = 169$
- 4 $OP = \sqrt{169} = 13$ cm

Answer: 13 cm

7

[3 marks]

Two angles at the circumference of a circle both stand on the same chord AB and are on the same side of it. One of them is 38° .

Write down the size of the other angle and give a reason.

- 1 Angles standing on the same chord, in the same segment, are equal.
- 2 So the other angle is also 38° .
- 3 This is the 'angles in the same segment' theorem.

Answer: 38° — angles in the same segment are equal

8

[3 marks]

A and B are points on a circle with centre O. Angle AOB = 70° .

Work out the size of angle OAB, explaining your reasoning.

- 1 OA and OB are both radii, so they are equal in length.
- 2 That makes triangle OAB isosceles, so angles OAB and OBA are equal.
- 3 The three angles total 180° , so the two equal angles share $180 - 70 = 110^\circ$.
- 4 $110 \div 2 = 55^\circ$

Answer: 55° — OA and OB are radii, so the triangle is isosceles

9

[4 marks]

A, B and C are points on a circle with centre O, with C on the major arc AB.

Angle OAB = 25° .

(a) Write down angle OBA, with a reason.

(b) Work out angle AOB.

(c) Work out angle ACB.

1 (a) OA and OB are radii, so triangle OAB is isosceles and angle OBA = 25° .

2 (b) The angles of triangle OAB total 180° .

3 Angle AOB = $180 - 25 - 25 = 130^\circ$

4 (c) The angle at the centre is twice the angle at the circumference on the same arc.

5 Angle ACB = $130 \div 2 = 65^\circ$

Answer: (a) 25° (isosceles, two radii) (b) 130° (c) 65°

10

[4 marks]

AB is a diameter of a circle with AB = 13 cm. C is a point on the circle with AC = 5 cm.

(a) Write down angle ACB, with a reason.

(b) Work out the length of BC.

(c) Work out the area of triangle ABC.

1 (a) Angle ACB = 90° , because the angle in a semicircle is a right angle.

2 (b) Triangle ACB is right-angled at C, so $BC^2 = 13^2 - 5^2$

3 = $169 - 25 = 144$, so BC = 12 cm

4 (c) AC and BC are perpendicular, so use them as base and height.

5 Area = $\frac{1}{2} \times 5 \times 12 = 30 \text{ cm}^2$

Answer: (a) 90° (angle in a semicircle) (b) 12 cm (c) 30 cm^2

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

The angle at the centre of a circle standing on an arc is 84° .

Work out the angle at the circumference standing on the same arc.

1 The angle at the centre is twice the angle at the circumference.

2 $84 \div 2 = 42^\circ$

Answer: 42°

2

[2 marks]

In a cyclic quadrilateral, two opposite angles are x° and $3x^\circ$.

Work out the value of x .

1 Opposite angles of a cyclic quadrilateral add up to 180° .

2 $x + 3x = 180$, so $4x = 180$

3 $x = 45$

Answer: $x = 45$

3

[3 marks]

PT is a tangent to a circle with centre O, touching at T. The radius is 6 cm and $PT = 8$ cm.

(a) Work out the length of OP.

(b) Work out the size of angle TPO, to 1 decimal place.

1 (a) Angle OTP = 90° because a tangent meets a radius at right angles.

2 $OP^2 = 6^2 + 8^2 = 100$, so $OP = 10$ cm

3 (b) At P, the opposite side is $OT = 6$ and the adjacent side is $PT = 8$.

4 $\tan(\text{TPO}) = \frac{6}{8} = 0.75$, so angle TPO = $\tan^{-1}(0.75) = 36.869\dots^\circ$

Answer: (a) 10 cm (b) 36.9°

4

[3 marks]

A tangent touches a circle at A. A chord AB is drawn, and the angle between the tangent and the chord is 54° .

Write down the size of the angle in the alternate segment, and state the theorem you have used.

1 The alternate segment theorem says the angle between a tangent and a chord equals the angle in the alternate segment.

2 The alternate segment is the one on the other side of the chord.

3 So the angle there is also 54° .

Answer: 54° — by the alternate segment theorem

5

[4 marks]

ABCD is a cyclic quadrilateral in which AB is a diameter of the circle.

Angle ADC = 130° .

(a) Work out angle ABC, with a reason.

(b) Write down angle ACB, with a reason.

(c) Work out angle BAC.

1 (a) ADC and ABC are opposite angles of a cyclic quadrilateral, so they add to 180° .

2 Angle ABC = $180 - 130 = 50^\circ$

3 (b) Angle ACB stands on the diameter AB, so it is 90° (angle in a semicircle).

4 (c) In triangle ABC the angles total 180° .

5 Angle BAC = $180 - 90 - 50 = 40^\circ$

Answer: (a) 50° (cyclic quadrilateral) (b) 90° (angle in a semicircle) (c) 40°

6

[4 marks]

A, B and C are points on a circle with centre O. Angle BAC = 35° .

(a) Work out angle BOC, with a reason.

(b) Work out angle OBC.

1 (a) The angle at the centre is twice the angle at the circumference on the same arc BC.

2 Angle BOC = $2 \times 35 = 70^\circ$

3 (b) OB and OC are radii, so triangle OBC is isosceles.

4 The two base angles share $180 - 70 = 110^\circ$.

5 Angle OBC = $110 \div 2 = 55^\circ$

Answer: (a) 70° (angle at centre is twice the angle at circumference) (b) 55°

7

[4 marks]

TA is a tangent to a circle, touching at A. ABCD is a cyclic quadrilateral.

Angle TAB = 65° and angle ABC = 80° .

(a) Work out angle ACB, with a reason.

(b) Write down angle ADB, with a reason.

(c) Work out angle ADC.

1 (a) Angle TAB is between the tangent and the chord AB, so it equals the angle in the alternate segment, which is angle ACB.

2 Angle ACB = 65°

3 (b) Angles ADB and ACB both stand on the chord AB in the same segment, so they are equal.

4 Angle ADB = 65°

5 (c) ABC and ADC are opposite angles of the cyclic quadrilateral.

6 Angle ADC = $180 - 80 = 100^\circ$

Answer: (a) 65° (alternate segment) (b) 65° (same segment) (c) 100°

8

[4 marks]

PT is a tangent to a circle with centre O, touching at T. The radius is 7 cm and OP = 25 cm.

(a) Work out the length of PT.

(b) Work out angle TOP, to 1 decimal place.

1 (a) Angle OTP = 90° , so $PT^2 = 25^2 - 7^2$

2 = $625 - 49 = 576$, so $PT = \sqrt{576} = 24$ cm

3 (b) At O, the adjacent side is OT = 7 and the hypotenuse is OP = 25.

4 $\cos(\text{TOP}) = \frac{7}{25} = 0.28$, so angle TOP = $\cos^{-1}(0.28) = 73.739\dots^\circ$

Answer: (a) 24 cm (b) 73.7°

9

[4 marks]

PA and PB are tangents to a circle with centre O, touching at A and B.

Angle APB = 48° .

(a) Write down angle OAP, with a reason.

(b) Work out angle AOB.

(c) Work out angle OAB.

1 (a) Angle OAP = 90° , because a tangent meets a radius at right angles.

2 (b) OAPB is a quadrilateral with angles 90° , 48° , 90° and AOB, totalling 360° .

3 Angle AOB = $360 - 90 - 90 - 48 = 132^\circ$

4 (c) OA and OB are radii, so triangle OAB is isosceles.

5 Angle OAB = $(180 - 132) \div 2 = 24^\circ$

Answer: (a) 90° (tangent meets radius at 90°) (b) 132° (c) 24°

10

[5 marks]

Prove that the angle at the centre of a circle is twice the angle at the circumference when both stand on the same arc.

1 Let O be the centre, A and B be the ends of the arc, and P a point on the major arc. Draw the line PO and extend it to meet the circle again at Q.

2 OA = OP because both are radii, so triangle OAP is isosceles and angle OAP = angle OPA. Call each of these a.

3 Angle AOQ is the exterior angle of triangle OAP at O, so angle AOQ = $a + a = 2a$.

4 In the same way OB = OP, so triangle OBP is isosceles with equal angles b, and angle BOQ = $2b$.

5 Angle AOB = angle AOQ + angle BOQ = $2a + 2b = 2(a + b)$

6 Angle APB = angle OPA + angle OPB = $a + b$, so angle AOB = $2 \times$ angle APB as required.

Answer: Splitting the figure with the line PO gives two isosceles triangles; each contributes an exterior angle of double its base angle, so the centre angle is $2(a + b) =$ twice the circumference angle